

Kevin Vaillancourt

Independent researcher — measurement and interpretability of language models

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RESEARCH

2026 –

IOTA — how much of a model's next output comes from the conversation, not the prompt

Independent, sole investigator · github.com/Vee-industries/Iota-Code

A collection protocol that holds the user prompt byte-identical at a given turn across every trial and temperature, so the history term can be varied on its own, and an estimator for what share of next-output predictability it carries. Three model families, five temperatures, one 10 GB consumer GPU.

- History carries 34–55% of next-output predictability in LLaMA-3-8B (lowest bootstrap bound +28%), 4–7% in Gemma-2-9B, and less again in Mistral-7B — three architectures, one protocol, three different answers.
- Showed the estimator's null is **negative** (–0.046 to –0.074 nats), not zero. Reporting against zero inverts the conclusion for two of the three families. The most transferable thing I have found.
- A system prompt instructing the model to disengage cuts history-dependence roughly fourfold on byte-identical questions — 46% to 12%, non-overlapping intervals.
- Retracted the project's original result after my own control falsified it: the three-channel decomposition of the next hidden state is zero by construction, and a no-history probe returned byte-identical states across cohorts that otherwise differ. I dropped the channel, redefined the measurand on the output, and re-established it there.
- Verified the surviving result transfers to open generation (31–49%, every interval overlapping the enforced corpus) — the apparatus's largest stated limitation, now tested.
- Apparatus, preregistered controls and corpus are public under Apache-2.0 and reproduce from a clone with no GPU.

2026

Detector Memory Glove — a stateful severity layer for intrusion detection

Apart Research / CeSIA AI Incident Response Sprint · github.com/Vee-industries/detector_memory_glove

- Per-entity exponential moving average scored beside the current event, on a synthetic testbed built to the published timing and phase structure of the July 2026 Hugging Face agent intrusion.
- At a 120-hour review window with false alarms held under 5%, detection rises from 13.3% [12.6, 13.9] without memory to 94.0% [93.6, 94.5] with it. 9,994 attack campaigns, 40,000 benign entities, bootstrap B = 2000.
- Includes a diagnostic locating predictive information in the event stream relative to an external risk label, and an account of when the memory advantage is real and when it is an artifact of warm-up.

PUBLICATIONS

- Apr 2026 **Formal Properties of the Structural Remainder R: An Information-Theoretic Measurement Target**
Zenodo · CC BY 4.0 · [10.5281/zenodo.19656944](https://doi.org/10.5281/zenodo.19656944)
Boundedness on $[0, 1]$, invariance under bijective reparameterisation, non-triviality via a closed-form one-bit cache model, identifiability by regime, and a closed form in stable linear-Gaussian systems. Scoped to systems with persistent internal state — state-space models and recurrent networks — and makes no claim about transformers.
- Apr 2026 **Decomposing Gait Robustness in Demonstration-Free Humanoid Locomotion**
Zenodo preprint · CC BY 4.0 · [10.5281/zenodo.19474028](https://doi.org/10.5281/zenodo.19474028)
Four preregistered diagnostic experiments with falsification criteria, applying the same decomposition to ROM-guided reinforcement learning for locomotion. Proposal; no results. Written for Dr. Meng Cheng Lau, Laurentian University, whose framework it builds on.

METHODS

- Estimation Conditional V-usable information from held-out log-loss, grouped cross-validation, cluster bootstrap, permutation nulls, Theil's U.
- Controls Preregistration with the decision rule fixed before the run, including arms that returned null and are reported as null. Length-matched scrambles, degradation ladders, and a bitwise reproducibility floor across independent model loads.
- Calibration A six-system synthetic suite with analytic ground truth, including a negative control the estimator failed and that I documented as a failure.
- Interventions Activation patching by forward hook into the residual stream with the emitted token as outcome; layer-isolation sweeps; ablated-versus-instruct contrasts.
- Tools Python, PyTorch, Transformers, NumPy, scikit-learn. 4-bit to FP16 quantisation via bitsandbytes; Hugging Face. Consumer-GPU inference under a hard 10 GB budget.
- Practice I specify the architecture, protocol and analysis and direct the implementation through AI tooling rather than typing it.

EMPLOYMENT AND EDUCATION

- 2022 – **Passport Officer, Service Canada**
Currently on medical leave · Secret clearance (Level II), CSIS security assessment, ten-year background review
- 2017 – 2022 **Canada Post**
- 2021 **BSc, Interdisciplinary Science — Laurentian University**
Quantitative methods, systems analysis, probability and statistics, scientific computing
- Languages English (native), French (working proficiency).